

ENME608: Mechanism and Dynamics of Machinery

Lecture 3: Linkage Position Analysis (Part 1)

Dr. Thein Than Tun

Lecturer – Mechatronics & Dynamics

School of Engineering, Computer and Mathematical Sciences



Assessment

Assessment Type	Weighting	Due Date
Mid-semester Test	25%	In Week 6 (Date & Time: TBA)
Assignment	25%	31 May 2026, Online Submission by 11:59 PM
Final Controlled Assessment	50%	After Week 12 (Date & Time: TBA)

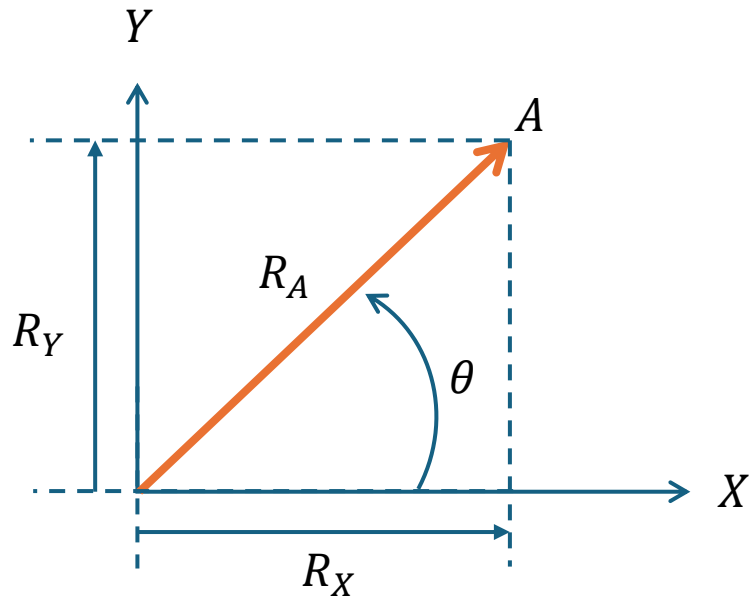
Contents of ENME608

Mid-Sem Break	Mechanism and Machine Concepts	Week 1	Contents of Mid-Semester Test
	Linkages	Week 2	
	Position Analysis	Week 3 – 4	
	Velocity Analysis	Week 5	
	Acceleration Analysis	Week 6	
	Cam	Week 7	Contents of Final Exam
	Cam (continued) and Gear	Week 8	
	Gear (continued)	Week 9	
	Dynamics Fundamentals	Week 10	
	Force Analysis	Week 11	
	Force Analysis (continued) and revision	Week 12	
Final Exam (TBA)			

Reference

- Recommended Textbook
 - Robert L. Norton, *Design of Machinery : An Introduction to the Synthesis and Analysis of Mechanisms and Machines*, 6th Edition, McGraw Hill, 2019, ISBN-10: 1260113310 , ISBN-13: 978-1260113310
(Online version available: AUT Library or
ENME608: Course Resources in Canvas)

Global Coordinate System (GCS)

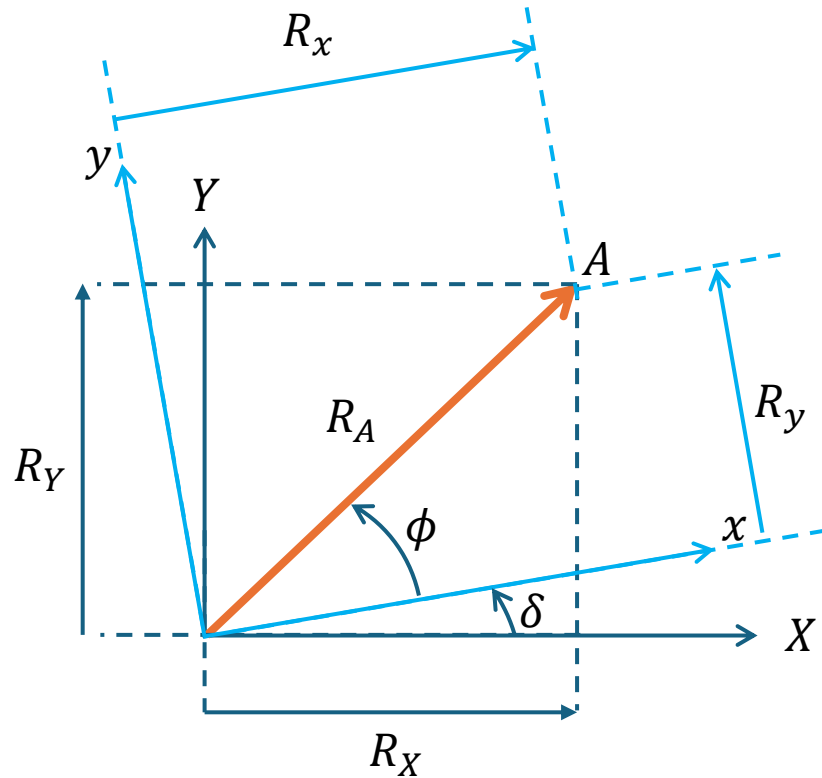


Polar form: $|R_A| @ \angle \theta$

Cartesian form: R_X, R_Y

$$R_A = \sqrt{R_X^2 + R_Y^2}$$
$$\theta = \arctan\left(\frac{R_Y}{R_X}\right)$$

Local Coordinate System (LCS)



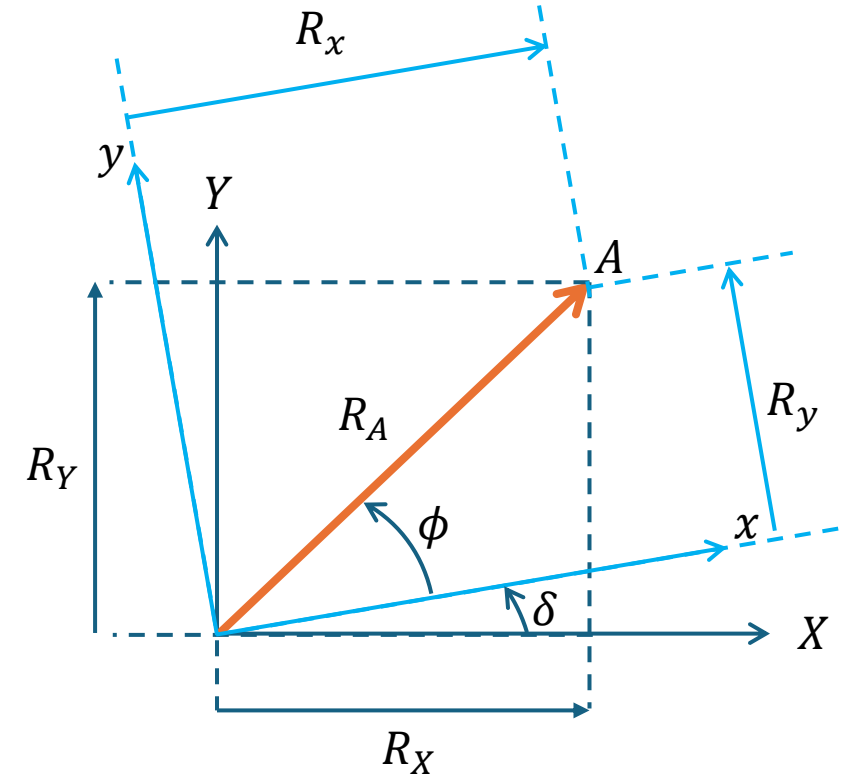
Polar form: $|R_A| @ \angle \phi$

Cartesian form: R_x, R_y

$$R_A = \sqrt{R_x^2 + R_y^2}$$
$$\phi = \arctan\left(\frac{R_y}{R_x}\right)$$

- Local Nonrotating Coordinate System (LNCS)
- Local Rotating Coordinate System (LRCS)

Coordinate Transformation



$$\begin{aligned} R_X &= R_x \cos \delta - R_y \sin \delta \\ R_Y &= R_x \sin \delta + R_y \cos \delta \end{aligned}$$

$\Leftrightarrow$

$$\begin{bmatrix} R_X \\ R_Y \end{bmatrix} = \begin{bmatrix} \cos \delta & -\sin \delta \\ \sin \delta & \cos \delta \end{bmatrix} \begin{bmatrix} R_x \\ R_y \end{bmatrix}$$

Vector Representation

Suppose $r = |R_A|$,

Polar form

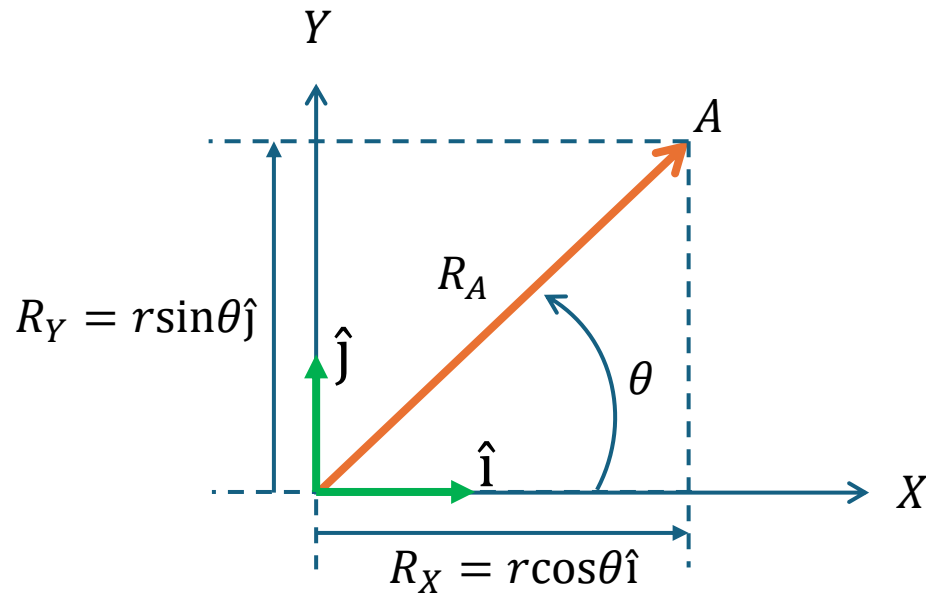
$r @ \angle \theta$

$re^{j\theta}$

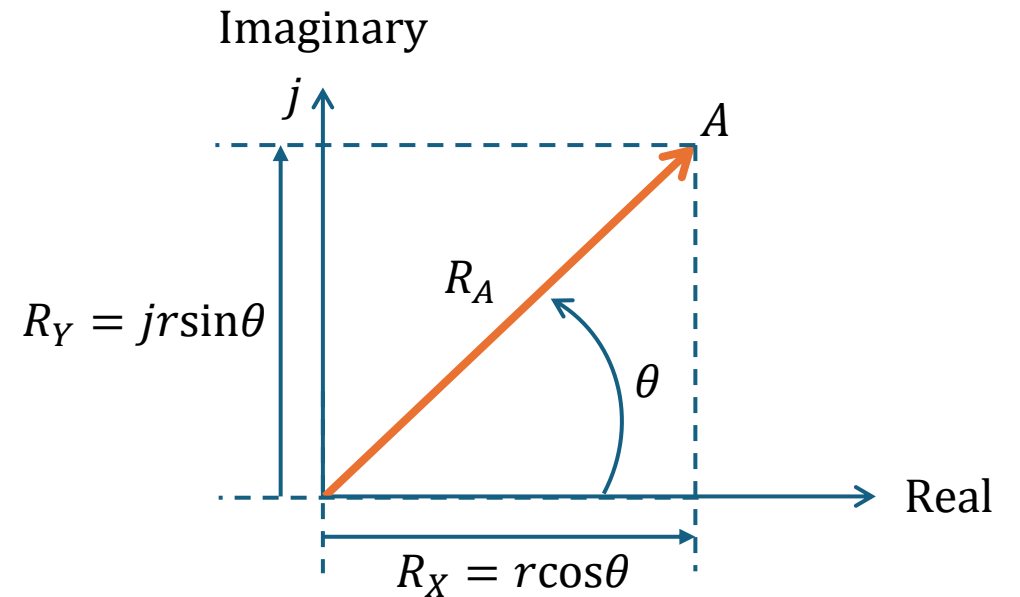
Cartesian form

$r\cos\theta\hat{i} + r\sin\theta\hat{j}$

$r\cos\theta + jrsin\theta$

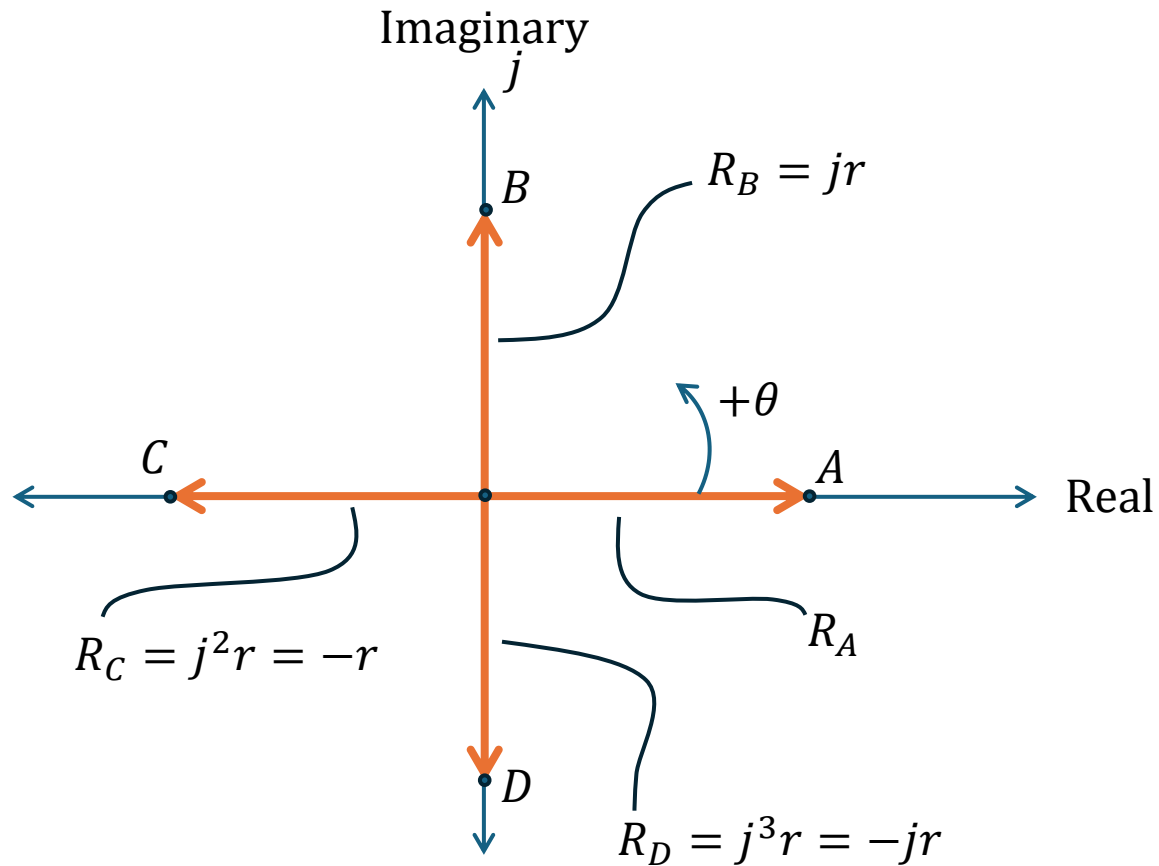


Unit vector notation



Complex number representation

Vector Rotation in Complex Plane



Suppose $r = |R_A|$,

Polar form

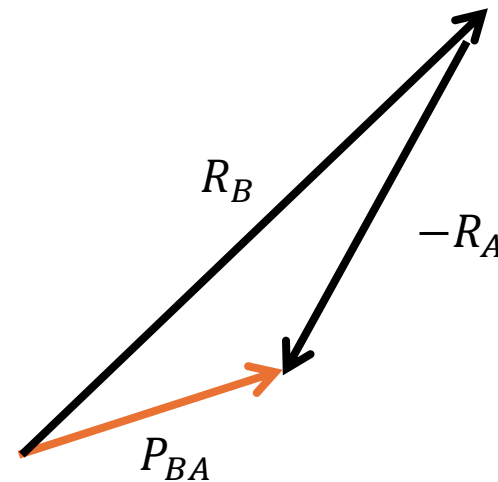
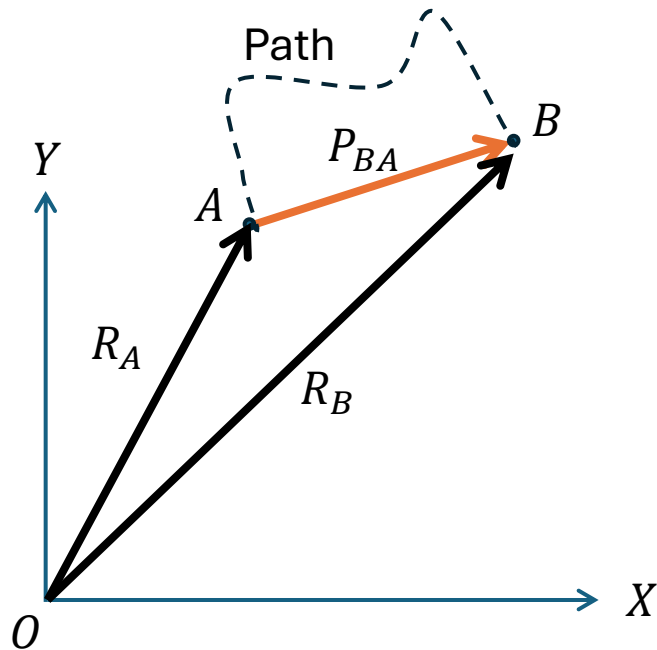
$$re^{j\theta}$$

Cartesian form

$$r\cos\theta + j r\sin\theta$$

Displacement (Linear)

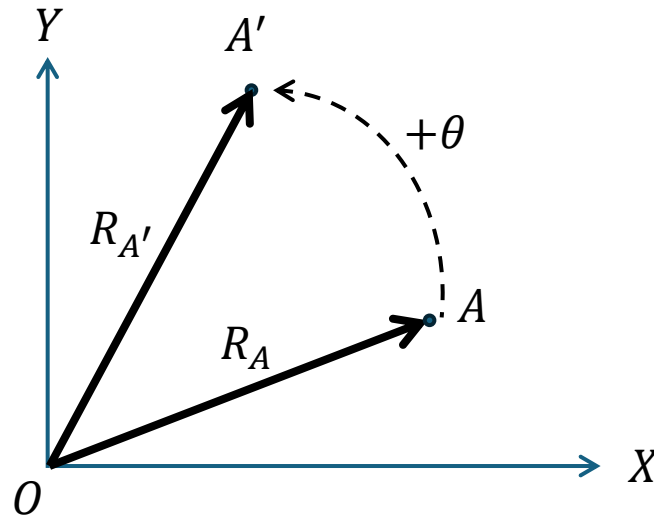
- the directional line (vector) from the initial to the final position of a point which has moved in the reference frame.
- Its rate of change with time is velocity (vector).



$$P_{BA} = R_B - R_A$$

Displacement (Angular)

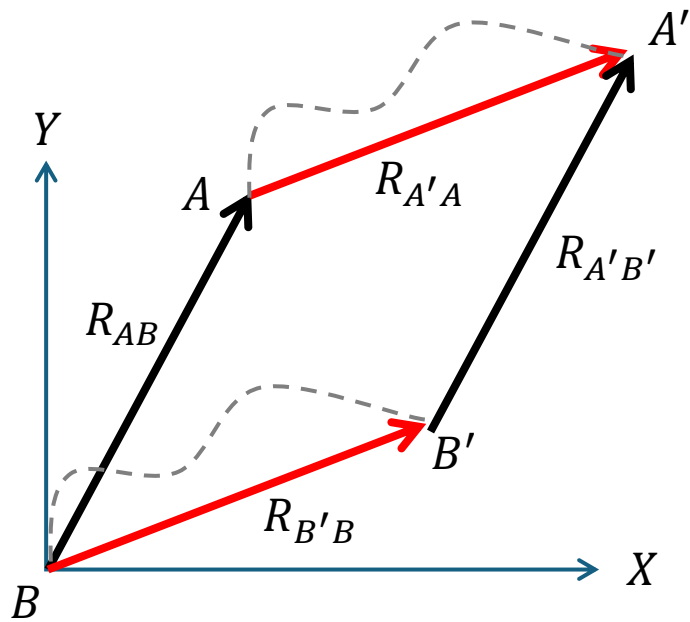
- amount of the change of angle of a rigid body.
- Its rate of change with time is angular velocity.



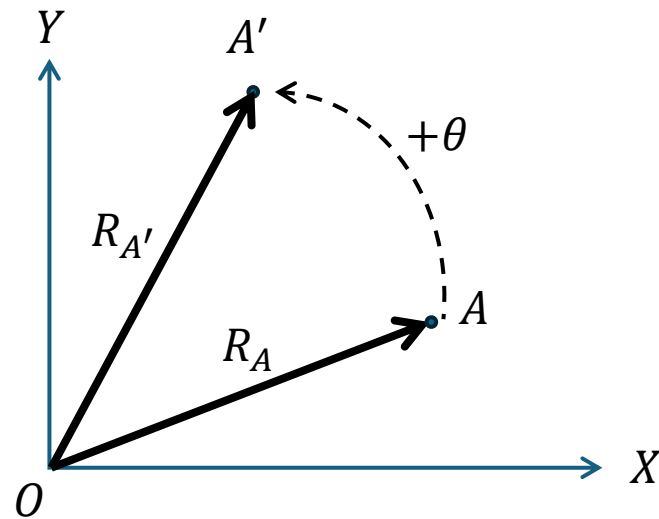
$$r = |R_A| = |R_{A'}|$$

Types of Motion of a Body

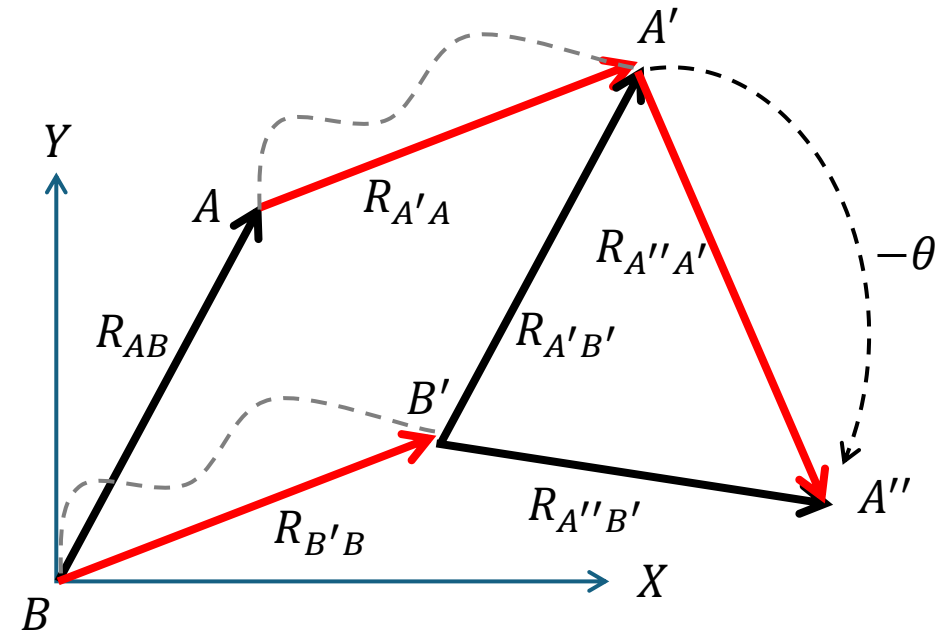
- _____ : Same displacement for all the points.
- _____ : Different displacement for different points while zero displacement for one point.
- _____ : Rotation + Translation



Curvilinear Translation

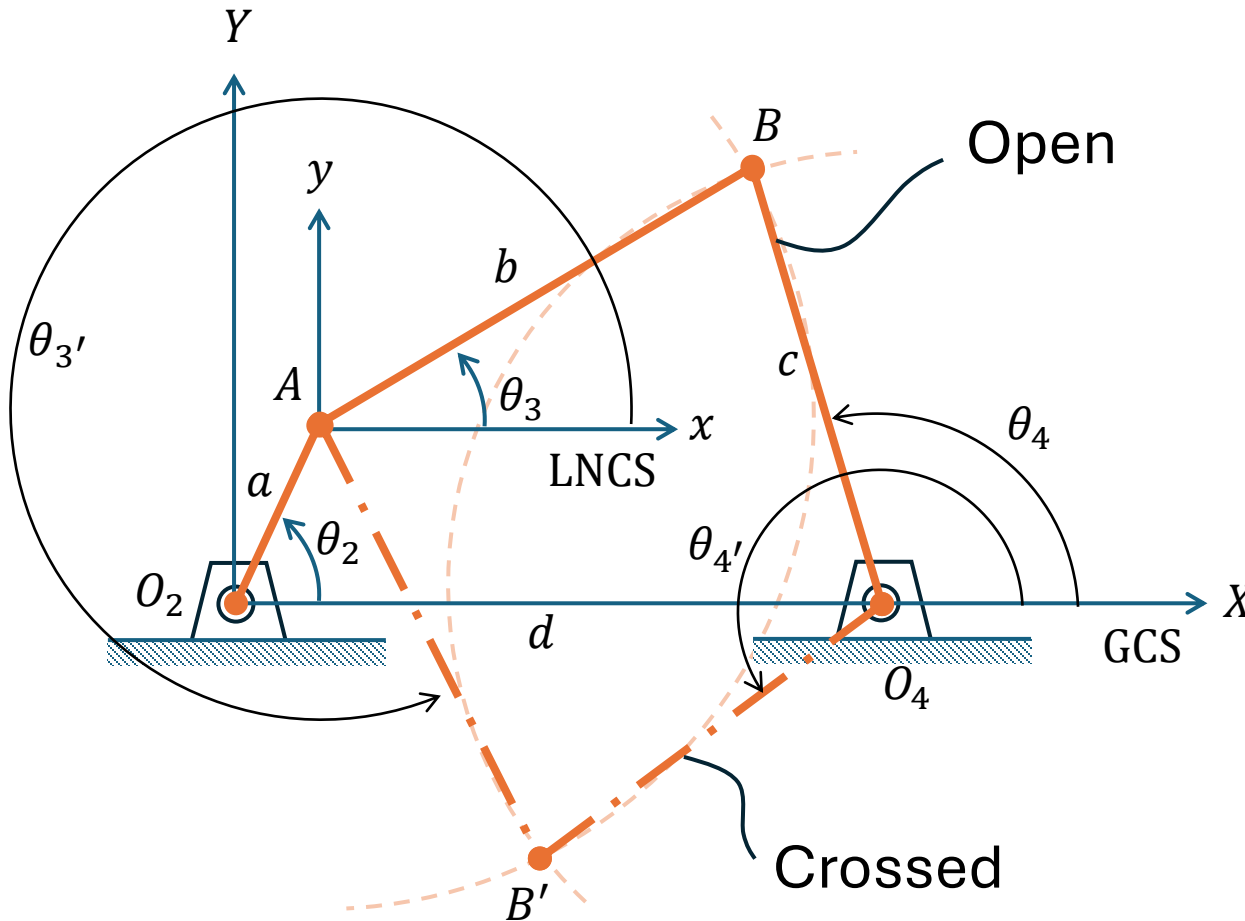


Rotation



Complex Motion

Position Analysis of Fourbar Linkage: Algebraic



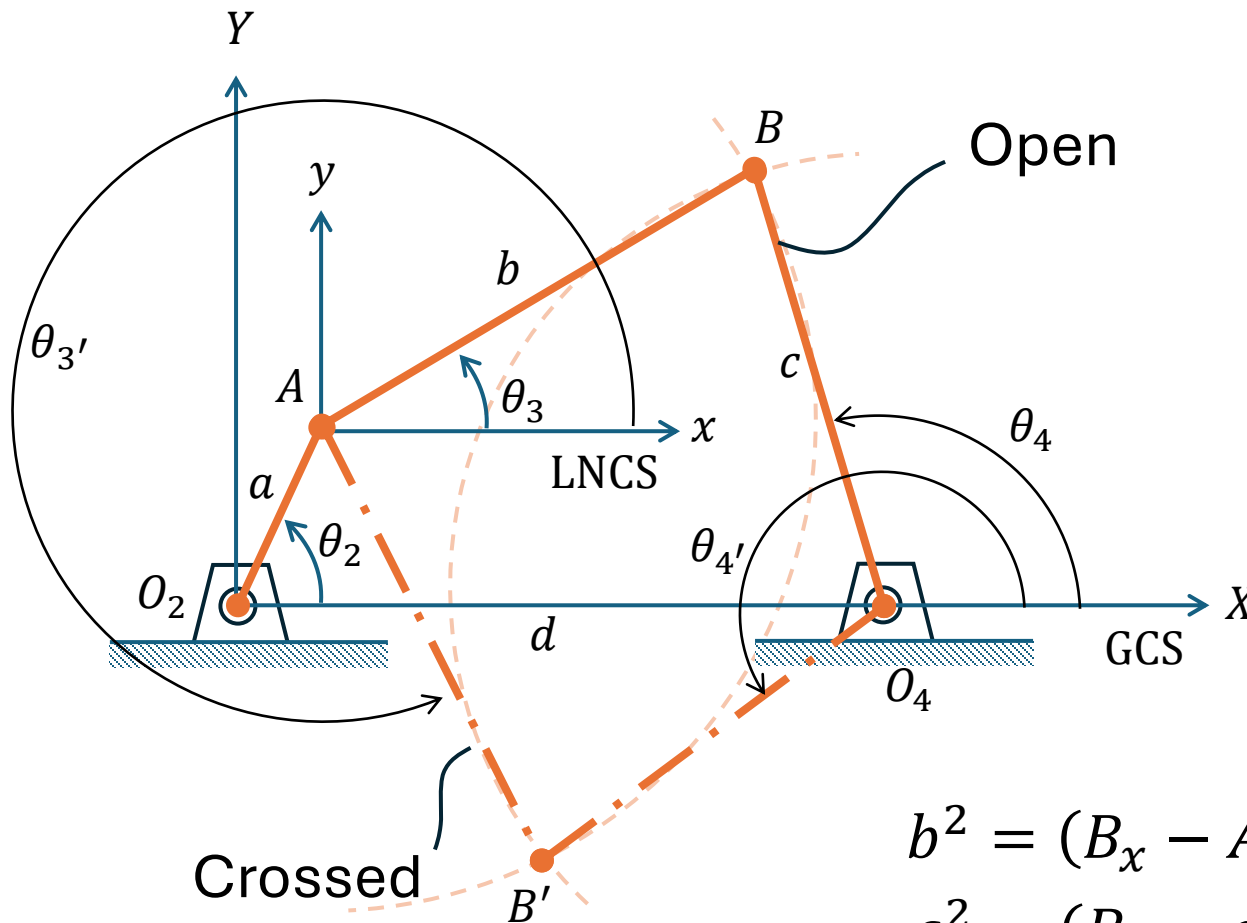
$$A_x = a \cos \theta_2$$

$$A_y = a \sin \theta_2$$

$$b^2 = (B_x - A_x)^2 + (B_y - A_y)^2$$

$$c^2 = (B_x - d)^2 + B_y^2$$

Position Analysis of Fourbar Linkage: Algebraic (Cont.)



Known Parameters: a, b, c, d, θ_2

Find: θ_3, θ_4

$$\left. \begin{aligned} A_x &= f(a, \theta_2) \\ A_y &= f(a, \theta_2) \end{aligned} \right\} \text{Known}$$

$$\left. \begin{aligned} b^2 &= (B_x - A_x)^2 + (B_y - A_y)^2 \\ c^2 &= (B_x - d)^2 + B_y^2 \end{aligned} \right\} \begin{array}{l} \text{Eqns.} \\ \text{Unknown} \end{array}$$

Position Analysis of Fourbar Linkage: Algebraic (Cont.)

$$B_x = S - \frac{2A_y B_y}{2(A_x - d)}$$

$$B_x = \frac{-Q + \sqrt{Q^2 - 4PR}}{2P}$$

where

$$Q = \frac{2A_y(d - S)}{A_x - d}$$

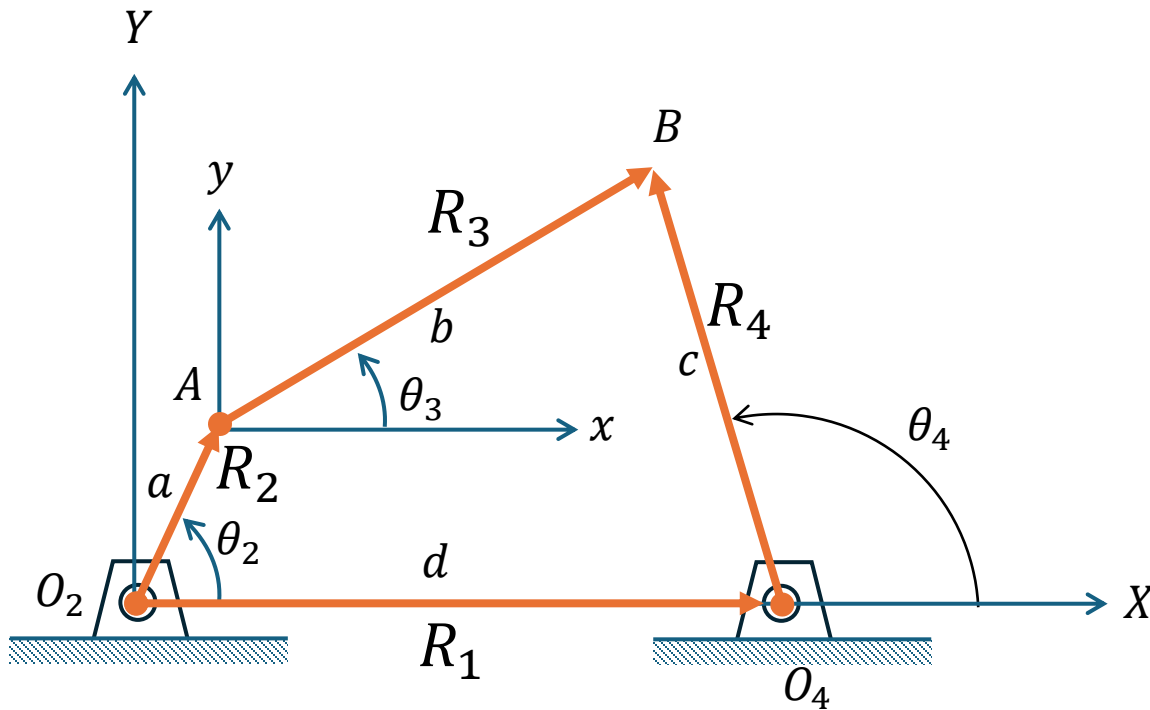
$$P = \frac{A_y^2}{(A_x - d)} + 1$$

$$R = (d - S)^2 - c^2 \quad S = \frac{a^2 - b^2 + c^2 - d^2}{2(A_x - d)}$$

$$\theta_3 = \tan^{-1} \left(\frac{B_y - A_y}{B_x - A_x} \right)$$

$$\theta_4 = \tan^{-1} \left(\frac{B_y}{B_x - d} \right)$$

Position Analysis of Fourbar Linkage: Vector Loop



Vectors	Alternate Notation	Link Length
R_2	R_A	a
R_3	R_{BA}	b
R_4	R_{BO_4}	c
R_1	R_{O_4}	d

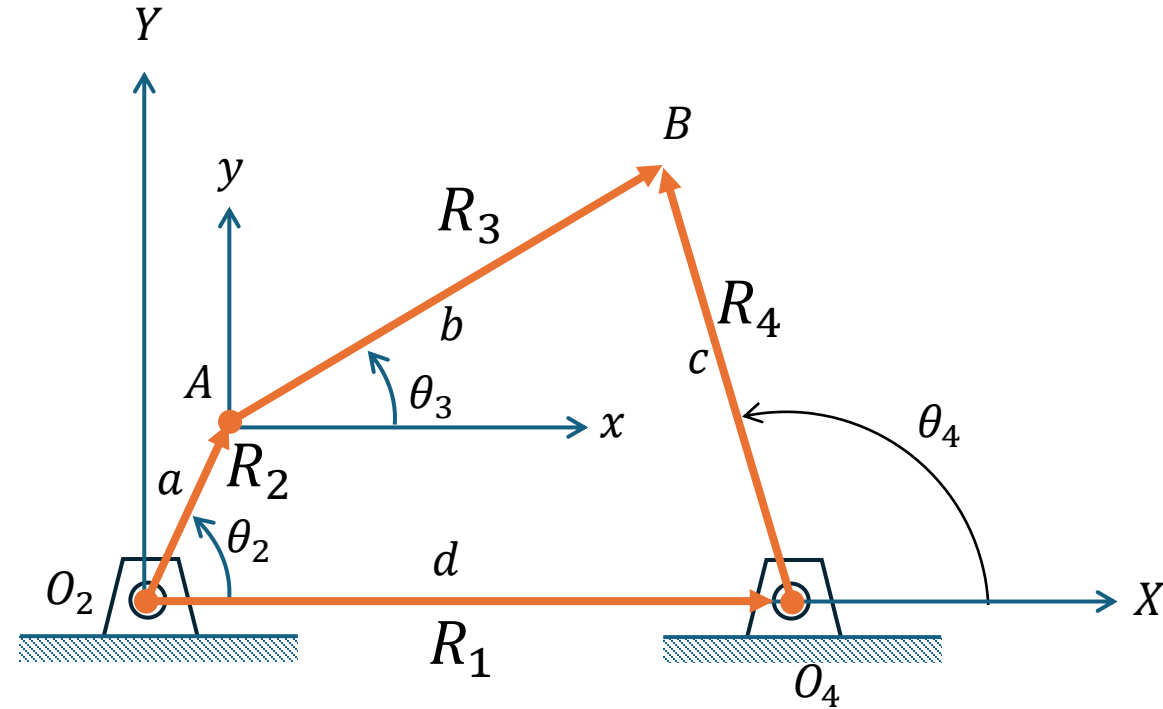
$$\text{Vector Loop: } R_2 + R_3 - R_4 - R_1 = 0$$

Position Analysis of Fourbar Linkage: Vector Loop (Cont.)

$$R_2 + R_3 - R_4 - R_1 = 0$$

$$ae^{j\theta_2} + be^{j\theta_3} - ce^{j\theta_4} - de^{j\theta_1} = 0$$

$\theta_1 = \text{_____}$



$$a(\cos \theta_2 + j \sin \theta_2) + b(\cos \theta_3 + j \sin \theta_3) - c(\cos \theta_4 + j \sin \theta_4) - d = 0$$

$$\text{Real : } a \cos \theta_2 + b \cos \theta_3 - c \cos \theta_4 - d = 0$$

$$\text{Imaginary : } a \sin \theta_2 + b \sin \theta_3 - c \sin \theta_4 = 0$$

_____ Eqns.
 _____ Unknown

Position Analysis of Fourbar Linkage: Vector Loop (Cont.)

$$\begin{aligned} a \cos \theta_2 + b \cos \theta_3 - c \cos \theta_4 - d &= 0 & \Leftrightarrow & b \cos \theta_3 = -a \cos \theta_2 + c \cos \theta_4 + d \\ a \sin \theta_2 + b \sin \theta_3 - c \sin \theta_4 &= 0 & \Leftrightarrow & b \sin \theta_3 = -a \sin \theta_2 + c \sin \theta_4 \end{aligned}$$

$$b^2 = a^2 + c^2 + d^2 - 2ad \cos \theta_2 + 2cd \cos \theta_4 - 2ac(\sin \theta_2 \sin \theta_4 + \cos \theta_2 \cos \theta_4)$$

$$\frac{b^2}{2ac} = \frac{a^2 + c^2 + d^2}{2ac} - \frac{2ad \cos \theta_2}{2ac} + \frac{2cd \cos \theta_4}{2ac} - \cos(\theta_2 - \theta_4)$$

$$\frac{a^2 - b^2 + c^2 + d^2}{2ac} - \frac{d \cos \theta_2}{c} + \frac{d \cos \theta_4}{a} = \cos(\theta_2 - \theta_4)$$

$$K_1 \cos \theta_4 - K_2 \cos \theta_2 + K_3 = \cos(\theta_2 - \theta_4) \quad \text{where} \quad K_1 = \frac{d}{a}, K_2 = \frac{d}{c}, K_3 = \frac{a^2 - b^2 + c^2 + d^2}{2ac}$$

Position Analysis of Fourbar Linkage: Vector Loop (Cont.)

Using half-angle identities

$$\sin\theta_4 = \frac{2\tan\left(\frac{\theta_4}{2}\right)}{1 + \tan^2\left(\frac{\theta_4}{2}\right)}, \quad \cos\theta_4 = \frac{1 - \tan^2\left(\frac{\theta_4}{2}\right)}{1 + \tan^2\left(\frac{\theta_4}{2}\right)}$$

$$K_1 \cos\theta_4 - K_2 \cos\theta_2 + K_3 = \cos(\theta_2 - \theta_4)$$

Simplification

$$A \tan^2\left(\frac{\theta_4}{2}\right) + B \tan\left(\frac{\theta_4}{2}\right) + C = 0$$

$$\theta_{4_{1,2}} = 2 \arctan\left(\frac{-B \pm \sqrt{B^2 - 4AC}}{2A}\right)$$

where $A = \cos\theta_2 - K_1 - K_2 \cos\theta_2 + K_3$

$$B = -2\sin\theta_2$$

$$C = K_1 - (K_2 + 1)\cos\theta_2 + K_3$$

Position Analysis of Fourbar Linkage: Vector Loop (Cont.)

Perform the same procedure on the following to solve θ_3

$$a \cos \theta_2 + b \cos \theta_3 - c \cos \theta_4 - d = 0$$

$$a \sin \theta_2 + b \sin \theta_3 - c \sin \theta_4 = 0$$

$$D \tan^2 \left(\frac{\theta_3}{2} \right) + E \tan \left(\frac{\theta_3}{2} \right) + F = 0$$

$$\theta_{3_{1,2}} = 2 \arctan \left(\frac{-E \pm \sqrt{E^2 - 4DF}}{2D} \right)$$

where $D = \cos \theta_2 - K_1 + K_4 \cos \theta_2 + K_5$

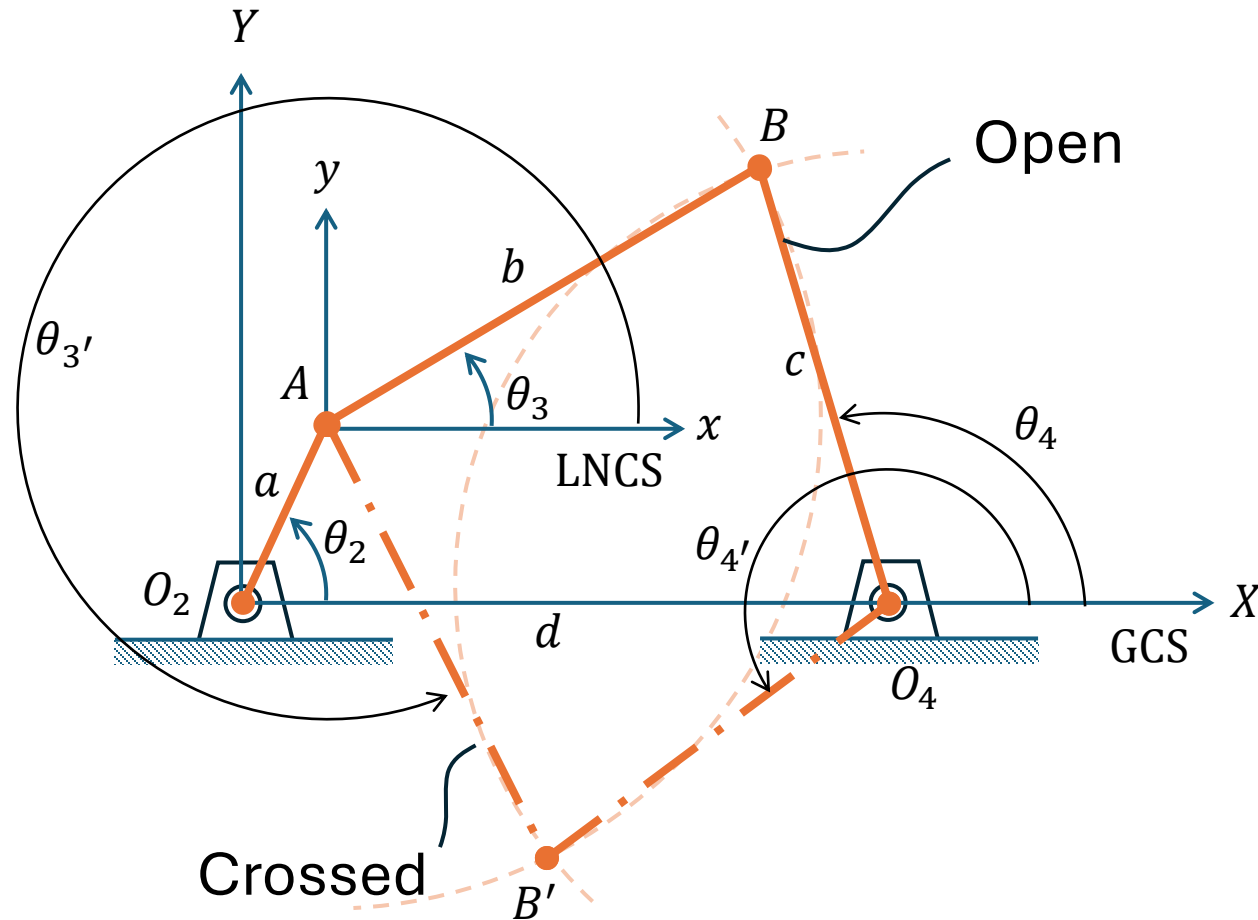
$$E = -2 \sin \theta_2$$

$$F = K_1 + (K_4 - 1) \cos \theta_2 + K_5$$

$$K_4 = \frac{d}{b}$$

$$K_5 = \frac{c^2 - d^2 - a^2 - b^2}{2ab}$$

Position Analysis of Fourbar Linkage: Example



Given a fourbar linkage with the link lengths

$$L_1 = d = 100\text{mm}$$

$$L_2 = a = 40\text{mm}$$

$$L_3 = b = 120\text{mm}$$

$$L_4 = c = 80\text{mm},$$

Find $\theta_2 = 40^\circ$, find all possible values of θ_3 and θ_4 .

Position Analysis of Fourbar Linkage: Example (Cont.)

$$K_1 = \frac{d}{a} = \frac{100}{40} = 2.5$$

$$K_2 = \frac{d}{c} = \frac{100}{80} = 1.25$$

$$K_3 = \frac{a^2 - b^2 + c^2 + d^2}{2ac} = \frac{40^2 - 120^2 + 80^2 + 100^2}{2(40)(80)} = 0.562$$

$$K_4 = \frac{d}{b} = \frac{100}{120} = 0.833$$

$$K_5 = \frac{c^2 - d^2 - a^2 - b^2}{2ab} = \frac{80^2 - 100^2 - 40^2 - 120^2}{2(40)(120)} = -2.042$$

$$A = \cos \theta_2 - K_1 - K_2 \cos \theta_2 + K_3 = \cos(40^\circ) - 2.5 - 1.25 \cos(40^\circ) + 0.562 = -2.129$$

$$B = -2 \sin \theta_2 = -2 \sin(40^\circ) = -1.286$$

$$C = K_1 - (K_2 + 1) \cos \theta_2 + K_3 = 2.5 - (1.25 + 1) \cos(40^\circ) + 0.562 = 1.339$$

$$D = \cos \theta_2 - K_1 + K_4 \cos \theta_2 + K_5 = \cos(40^\circ) - 2.5 + 0.833 \cos(40^\circ) - 2.042 = -3.137$$

$$E = -2 \sin \theta_2 = -2 \sin(40^\circ) = -1.286$$

$$F = K_1 + (K_4 - 1) \cos \theta_2 + K_5 = 2.5 + (0.833 - 1) \cos(40^\circ) - 2.042 = 0.331$$

Position Analysis of Fourbar Linkage: Example (Cont.)

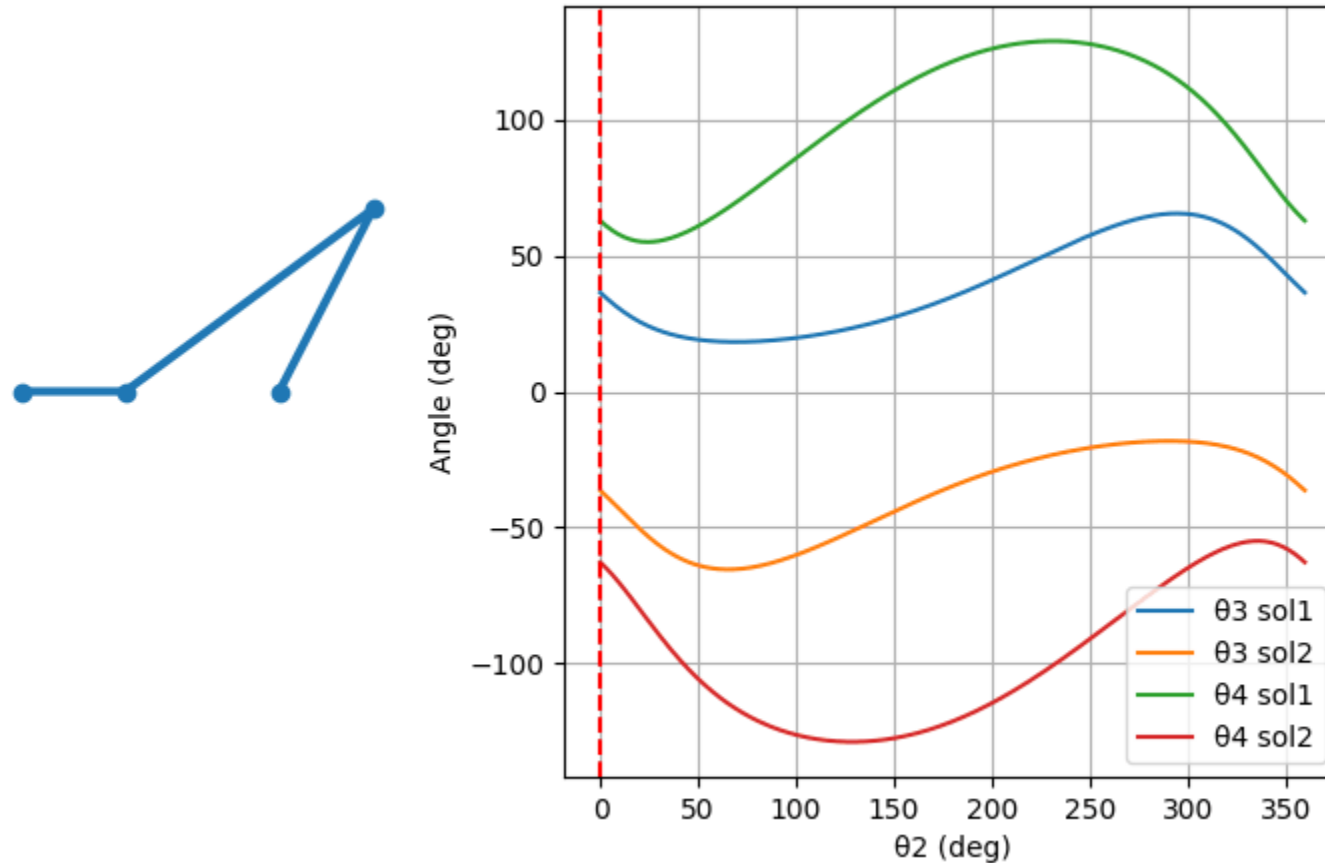
$$\theta_{4_{open}} = 2 \arctan \left(\frac{-B - \sqrt{B^2 - 4AC}}{2A} \right) = 2 \arctan \left(\frac{1.286 - \sqrt{-1.286^2 - 4(-2.129)(1.339)}}{2(-2.129)} \right)$$
$$= 57.33^\circ$$

$$\theta_{4_{crossed}} = 2 \arctan \left(\frac{-B + \sqrt{B^2 - 4AC}}{2A} \right) = 2 \arctan \left(\frac{1.286 + \sqrt{-1.286^2 - 4(-2.129)(1.339)}}{2(-2.129)} \right)$$
$$= -98.01^\circ$$

$$\theta_{3_{open}} = 2 \arctan \left(\frac{-E - \sqrt{E^2 - 4DF}}{2D} \right) = 2 \arctan \left(\frac{1.286 - \sqrt{-1.286^2 - 4(-3.137)(0.331)}}{2(-3.137)} \right)$$
$$= 20.30^\circ$$

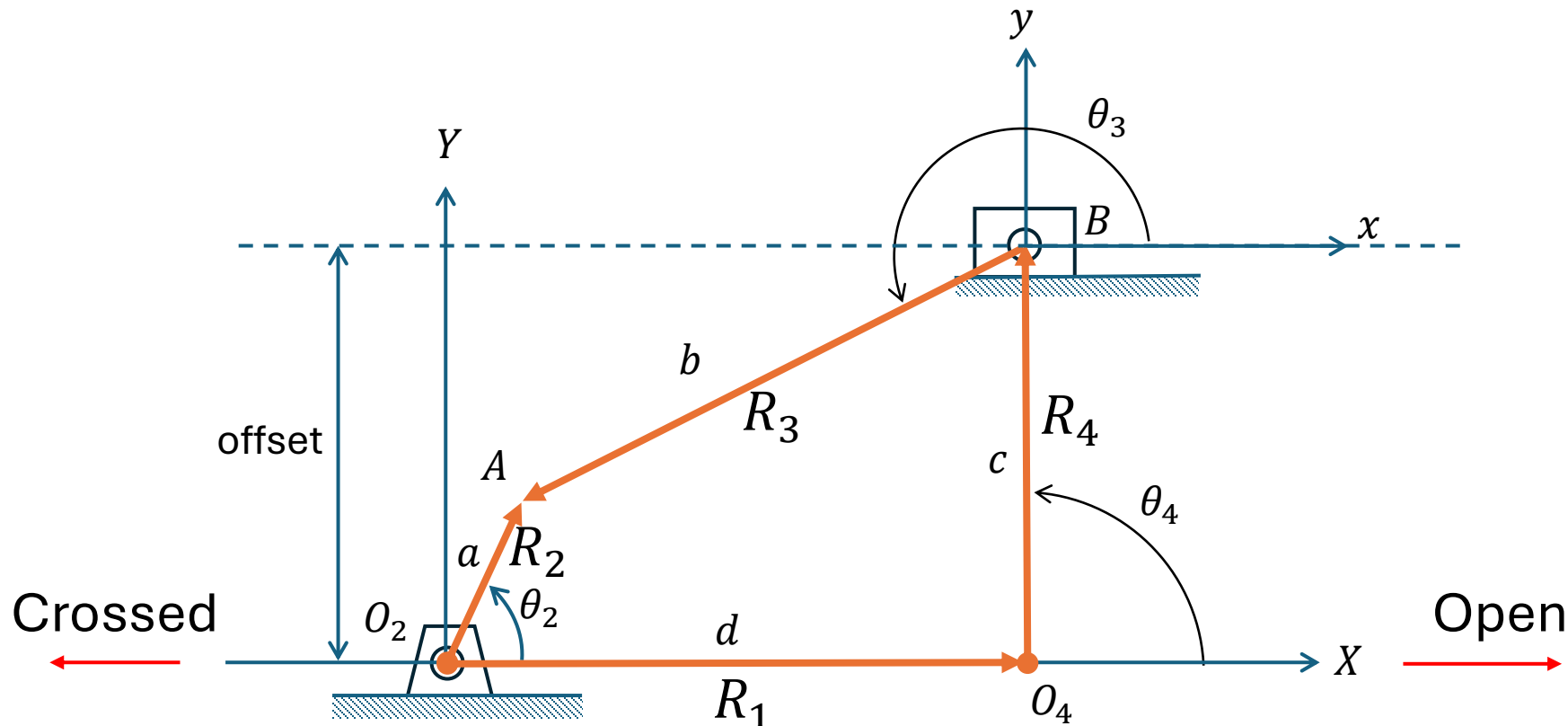
$$\theta_{3_{crossed}} = 2 \arctan \left(\frac{-E + \sqrt{E^2 - 4DF}}{2D} \right) = 2 \arctan \left(\frac{1.286 + \sqrt{-1.286^2 - 4(-3.137)(0.331)}}{2(-3.137)} \right)$$
$$= -60.98^\circ$$

Position Analysis of Fourbar Linkage: Example (Cont.)



Position Analysis of Crank-Slider: Vector Loop

$$\text{Vector Loop: } R_2 - R_3 - R_4 - R_1 = 0$$



Vectors	Link Length
R_2	a
R_3	b
R_4	c
R_1	d

Position Analysis of Crank-Slider: Vector Loop (Cont.)

$$R_2 - R_3 - R_4 - R_1 = 0$$

$$ae^{j\theta_2} - be^{j\theta_3} - ce^{j\theta_4} - de^{j\theta_1} = 0$$

$\theta_4 = \underline{\hspace{2cm}}$
 $\theta_1 = \underline{\hspace{2cm}}$

$$a(\cos \theta_2 + j \sin \theta_2) - b(\cos \theta_3 + j \sin \theta_3) - jc - d = 0$$

$$\text{Real : } a \cos \theta_2 - b \cos \theta_3 - d = 0$$

$$\text{Imaginary : } a \sin \theta_2 - b \sin \theta_3 - c = 0$$

} $\underline{\hspace{2cm}}$ **Eqns.**
 $\underline{\hspace{2cm}}$ **Unknown**

$$\theta_3 = \arcsin\left(\frac{a \sin \theta_2 - c}{b}\right), \arcsin\left(-\frac{a \sin \theta_2 - c}{b}\right) + \pi$$

$$d = a \cos \theta_2 - b \cos \theta_3$$

Position Analysis of Crank-Slider: Example

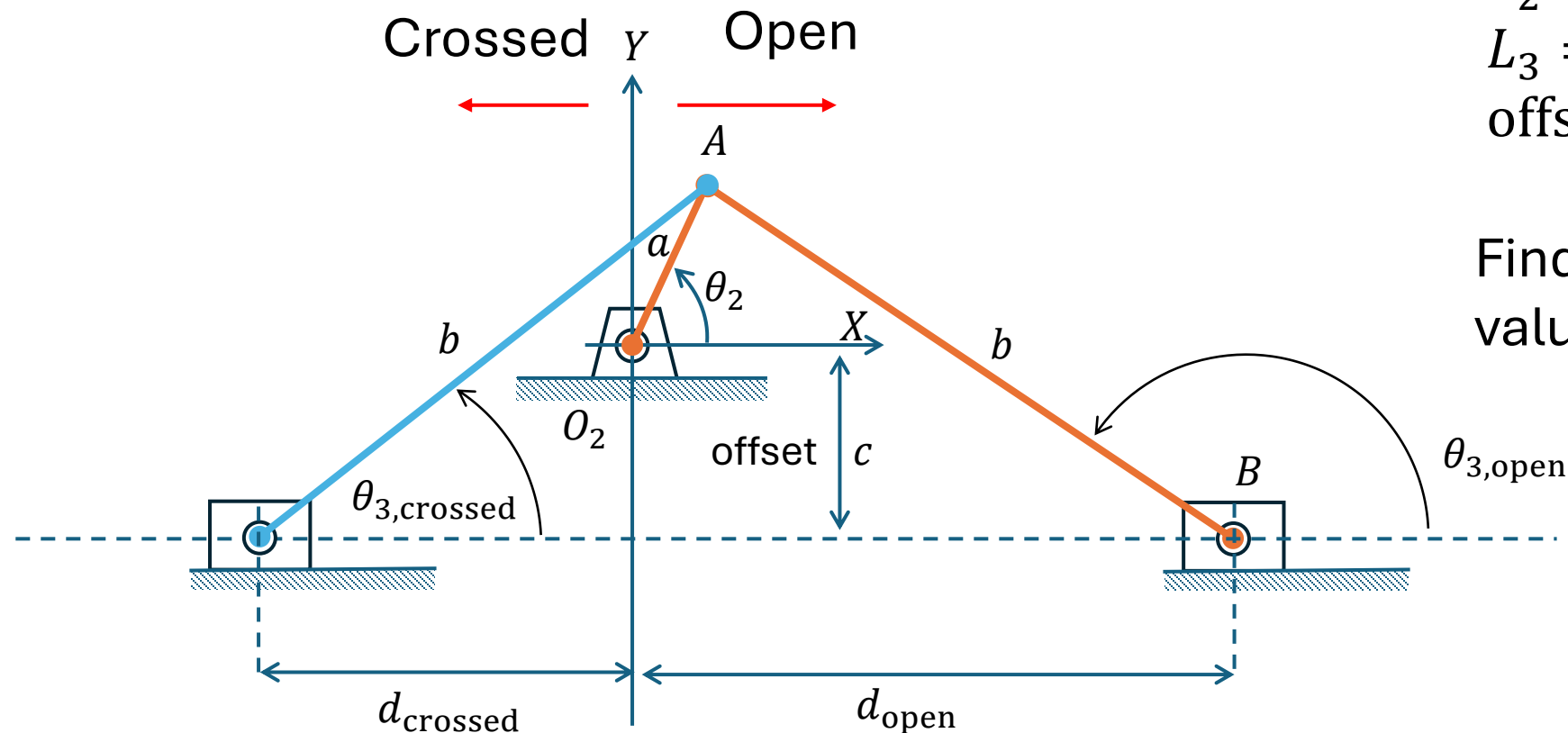
Given a fourbar linkage with the link lengths

$$L_2 = a = 40\text{mm}$$

$$L_3 = b = 120\text{mm}$$

$$\text{offset} = c = -20\text{mm},$$

Find $\theta_2 = 60^\circ$, find all possible values of θ_3 and d .



Position Analysis of Crank-Slider: Example

$$\theta_{3_{open}} = \arcsin\left(\frac{a \sin \theta_2 - c}{b}\right) = \arcsin\left(\frac{40 \sin(60^\circ) - (-20)}{120}\right) = 152.91^\circ$$

$$d = a \cos \theta_2 - b \cos \theta_3 = 40 \cos(60^\circ) - 120 \cos(152.91^\circ) = 126.84 \text{ mm}$$

$$\theta_{3_{crossed}} = \arcsin\left(-\frac{a \sin \theta_2 - c}{b}\right) + \pi = \arcsin\left(-\frac{40 \sin(60^\circ) - (-20)}{120}\right) + \pi = 27.09^\circ$$

$$d = a \cos \theta_2 - b \cos \theta_3 = 40 \cos(60^\circ) - 120 \cos(27.09^\circ) = -86.84 \text{ mm}$$

Position Analysis of Crank-Slider: Example (Cont.)

